## Problem Statements

1) A travel booking system integrates multiple airline APIs. Each API uses different methods such as `getFlightDetails()`, `fetchFlights()`, and `retrieveFlightInfo()`, while the application expects a standard method `searchFlights()`. Developers introduce a component that converts these different method calls into one unified method. Identify and implement the pattern and explain how it helps integrate heterogeneous systems.

2). An image gallery application loads large images using methods like `loadImage()` and `displayImage()`. Loading images consumes significant time, especially when repeatedly viewing the same image. Developers want to introduce a mechanism that delays loading until required and avoids duplicate loading. Identify and implement the pattern and explain its role in optimization.

3) A map application displays thousands of identical markers such as restaurants, hospitals, and ATMs. Each marker stores similar data like icon and color but differs in location. Developers want to minimize memory usage by sharing common marker data. Identify and implement the pattern and explain how intrinsic and extrinsic states are managed.

4) A university portal requires students to interact with multiple services such as `registerCourse()`, `payFees()`, `viewResults()`, and `applyCertificate()`. Currently, each service must be accessed individually. Developers want to provide a unified interface like `studentPortalAccess()` that simplifies interactions. Identify and implement the pattern and explain how it improves usability.